

STATEMENT OF WORK

Low Pressure Air Compressor [LPAC]

1. **SCOPE.**

- 1.1. Portsmouth Naval Shipyard (PNSY) is seeking to lease electric powered Low Pressure Air Compressor (LPAC) systems with support services (Start-up/commissioning, training, preventative maintenance and weekly equipment inspections) for use at Sub Base New London, CT. Quantities are listed in paragraph 3.1 component list. PNSY will use the LPAC Systems to establish a Breathing Air System, Tooling Air System and Blast Air System. The minimum allowable requirements for the systems are detailed in this Statement of Work. Unless otherwise stated, the contractor shall furnish the Low Pressure Air Compressor systems, including all labor, supervision, tools, materials, equipment and other items and services that are necessary to perform as described within this Statement of Work.

2. **APPLICABLE DOCUMENTS.**

- 2.1. **Issues of Publications.** The following documents in effect on the date of this requested proposal is a part of this specification to the extent applicable, or as specified herein. Within the specification, they shall be referred to by their basic designation only.

2.1.1. **Government Furnished**

2.1.1.1. N/A

2.1.2. **Nongovernment Furnished**

2.1.2.1. CGA G-7.1-2018: Commodity Specification for Air, Seventh Edition

2.1.2.2. OSHA 29 CFR 1910.134 – Respiratory Protection

2.1.2.3. ISO 8573-1:2001

2.1.2.4. OSHA 29 CFR 1910.169, Compressed Gas and Compressed Air Equipment, Air Receivers

- 2.2. **Order of precedence.** In the event of conflict between the Statement of Work (SOW) and the references cited above, the text of this SOW shall take precedence. Nothing in this SOW, however, shall supersede applicable laws or regulations unless a specific exemption has been obtained.

3. **REQUIREMENTS.**

3.1. **Component List.**

- 3.1.1. 100 HP LPAC Required components (Breathing Air System).

PIECE NO.	QTY.	NAME OF PART	PARA NO.
PC-01	1	100 HP LPAC (400 CFM)	3.2 thru 3.7
PC-02	1	500 Gal Air Receiving Tank	3.6
PC-03	1	400 CFM Desiccant Air Dryer	3.7
PC-04	250 FT	3 IN Air Hose	3.21
PC-05	1	E-stop and cabling	3.2.11

- 3.1.2. 200 HP LPAC Required components (Tooling Air System).

PIECE NO.	QTY.	NAME OF PART	PARA NO.
PC-06	1	200 HP LPAC (800 CFM)	3.8 thru 3.13
PC-07	1	1000 Gal Air Receiving Tank	3.12
PC-08	1	800 CFM Desiccant Air Dryer	3.13
PC-09	250 FT	3 IN Air Hose	3.21
PC-10	1	E-Stop and cabling	3.8.11

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3.1.3. 400 HP LPAC Required Components (Blasting Air System).

PIECE NO.	QTY.	NAME OF PART	PARA NO.
PC-11 thru PC-13	3	400 HP LPAC (1600 CFM)	3.14
PC-14 thru PC-16	3	1000 Gal Air Receiving Tank	3.18
PC-17 thru PC-19	3	1600 CFM Desiccant Air Dryer	3.19
PC-20	500 FT	3 IN Air Hose	3.21
PC-21	500 FT	3 IN Air Hose	3.21
PC-22	400 FT	2 IN Air Hose	3.21
PC-23	500 FT	3 IN Air Hose	3.21
PC-24 thru PC-26	3	E-Stop and cabling	3.14.11
CP-27 thru PC 31	5	Compressed Air Manifold	3.20

3.2. **100 HP Electric LPAC Requirements-[Breathing Air System].**

- 3.2.1. Nominal Compressed air flow rate: 400 SCFM.
- 3.2.2. Maximum system pressure: 125 PSIG.
- 3.2.3. Regulated delivered Air Pressure Set point: 115 PSIG.
- 3.2.4. Compressor shall be designed to deliver 100% Oil Free air capable of meeting the requirements as defined in section(s) 2.1.2.1 thru 2.1.2.4. Oil Flooded compressors are not acceptable.
- 3.2.5. Compressor connection: 3 Inch. Compressor must be supplied with Male/Female SPUD Fitting(s) as required for connection to downstream drying/filtering equipment.
- 3.2.6. Compressor shall be a fully packaged Air-Cooled unit including intercooler/after cooler to maintain a package maximum discharge compressed air temperature of 100F under all ambient conditions in order to protect downstream drying/filtering equipment.
- 3.2.7. Compressor shall enclosed in a weatherproof enclosure suitable for outdoor operation with an ambient operating temperature range of -10°F to 110°F without the need for additional shelter/protection.
- 3.2.8. All opening/connections/flanges are to be cleaned and capped.
- 3.2.9. Compressor shall be UL listed.
- 3.2.10. All vendor supplied equipment shall be capable of operating in below freezing temperatures and include freeze protection measures on components that are prone to freezing (e.g. condensate line, water separators, and drains)
- 3.2.11. Breathing Air LPAC must be pre-wired for and compatible with a Remote Emergency Stop (E-stop) Switch. This will enable the LPAC to be shut down in response to an emergency condition where breathing air needs to be secured. The Contractor shall route all E-stops switch & control wiring to a central location designated by the government where the breathing air LPAC equipment can be secured.
 - 3.2.11.1. The E-stop cable distance to be routed is estimated at 500 Ft, however, the Contractor shall be prepared to provide & route up to 1500 FT of wire to accommodate complex routing paths, cable routing around obstacles, service brows, electrical service chases, etc.
 - 3.2.11.2. E-stop switch shall be equipped with protective Plexiglas or acrylic style clear cover to prevent inadvertent activation.
 - 3.2.11.3. E-stop cabling must utilize heavy-duty industrial cable for use in flexible, portable, and extra-hard usage applications (type SO cable or equivalent).

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3.3. 100 HP LPAC Physical Dimension

- 3.3.1. Maximum Allowable Footprint (L x W) Inches: 90 X 54
- 3.3.2. Maximum Allowable Weight (lbs.): N/A.

3.4. 100 HP LPAC Electrical Requirements:

- 3.4.1. Operating Voltage: 460 VAC 3 Phase 60 Hz
- 3.4.2. Maximum Starting Amperage: 400A
- 3.4.3. Maximum Operating Amperage (FLA): 200A
- 3.4.4. All onboard low voltage controls must have integral transformer to provide necessary voltage stepdown. Compressor must operate when supplied from one 460VAC circuit.
- 3.4.5. Equipment shall come with Camlok connections on the unit (J series locking style, i.e.:1016 series)

3.5. 100 HP LPAC Operating and Safety Controls:

- 3.5.1. Compressor shall cycle using control logic based on maintaining system pressure.
- 3.5.2. Compressor and any pressure vessels within the package must be fitted with appropriate safety relief valve(s).
- 3.5.3. Pressure relief set point shall not exceed the working pressure rating of any component in line with the compressed air system (125 PSI).
- 3.5.4. Alarm(s)/controls to maintain compressor operable while meeting the requirements of Grade D Breathing Air and all compressed air quality requirements as defined in this statement of work.
- 3.5.5. The Breathing Air Compressor shall be equipped with Remote Emergency Stop Switch E-stop as defined in paragraph 3.2.11 of this SOW.

3.6. 500 Gallon Compressed Air Receiver Tank Requirements:

- 3.6.1. 500 Gallon, ASME rated Pressure Vessel
- 3.6.2. Minimum working pressure rating: 150 PSIG
- 3.6.3. Pressure Relief valve and tank drain meeting the pressure requirements to protect the supplied air receiver and compressed air system.
- 3.6.4. Tank maintenance & construction shall meet all requirements in accordance with Section 2.1.2.4
- 3.6.5. Vertical mounting Orientation.
- 3.6.6. Tank shall be equipped with lashing/ tie down points and base mount bolt pattern to enable the tank to be properly secured.
- 3.6.7. 3 IN connections: SPUD Fittings, M & F
- 3.6.8. Air Receiving tank shall be 100% free of any oily residue and shall be suitable to support and meet all requirements defined in section 2.1.2.

3.7. Desiccant Dryer Requirements:

- 3.7.1. Nominal Inlet Flow rate: 400 SCFM
- 3.7.2. Compressed air outlet flow Dew Point: -40°F PDP under full load of the supplied breathing air compressor.
- 3.7.3. Ambient Operating Temperature range: -10°F to 110°F
- 3.7.4. Maximum Allowable Footprint (L x W) Inches: 92 X 42
- 3.7.5. Maximum Allowable Weight (lbs.): N/A
- 3.7.6. Nominal Rated Working Pressure (PSIG): 150
- 3.7.7. Inlet / Outlet Connections: 3IN NPT
 - 3.7.7.1. Note: If the air dryer is manufactured with 2 IN NPT connections, adaptors shall be required to enable 3 IN SPUD connections for hose connections.
 - 3.7.7.2. NPT to 3 IN SPUD adaptors.
- 3.7.8. Dryer shall be equipped with particulate and oil removal pre-filters c/w automatic condensate drains and a one micron dust after filter.

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- 3.7.9. The desiccant dryer shall consume no more than 18% of the LPA input volume for purging to a – 40°F dew Point.
- 3.7.10. The Desiccant dryer shall operate from one circuit rated at: 120 VAC, single phase, 20 Amp, 60Hz.
- 3.7.11. Dryer shall be UL listed.

3.8. **200 HP Electric LPAC Requirements-[Tooling Air System].**

- 3.8.1. Nominal Compressed air flow rate: 800 SCFM.
- 3.8.2. Maximum system pressure: 125 PSIG.
- 3.8.3. Regulated delivered Air Pressure Set point: 115 PSIG.
- 3.8.4. Compressor shall be designed to deliver 100% Oil Free air. Oil Flooded compressors are not acceptable.
- 3.8.5. Compressor connection: 3 Inch. Compressor must be supplied with Male/Female SPUD Fitting(s) as required for connection to downstream drying/filtering equipment.
- 3.8.6. Compressor shall be a fully packaged Air-Cooled unit including intercooler/after cooler to maintain a package maximum discharge compressed air temperature of 100F under all ambient conditions.
- 3.8.7. Compressor shall enclosed in a weatherproof enclosure suitable for outdoor operation with an ambient operating temperature range of -10°F to 110°F without the need for additional shelter/protection.
- 3.8.8. All opening/connections/flanges are to be cleaned and capped.
- 3.8.9. Compressor shall be UL listed.
- 3.8.10. All vendor supplied equipment shall be capable of operating in below freezing temperatures and include freeze protection measures on components that are prone to freezing (e.g. condensate line, water separators, and drains)
- 3.8.11. Tooling Air LPAC must be pre-wired for and compatible with a Remote Emergency Stop (E-stop) Switch. This will enable the LPAC to be shut down in response to an emergency condition where tooling air needs to be secured. The Contractor shall route all E-stops switch & control wiring to a central location designated by the government where the breathing air LPAC equipment can be secured.
 - 3.8.11.1. The e-stop cable distance to be routed is estimated at 500 Ft, however, the Contractor shall be prepared to provide & route up to 1500 FT of wire to accommodate complex routing paths, cable routing around obstacles, service brows, electrical service chases, etc.
 - 3.8.11.2. E-stop switch shall be equipped with protective Plexiglas or acrylic style clear cover to prevent inadvertent activation.
 - 3.8.11.3. E-stop cabling must utilize heavy-duty industrial cable for use in flexible, portable, and extra-hard usage applications (type SO cable or equivalent).
- 3.8.12. All opening/connections/flanges are to be cleaned and capped.

3.9. **200 HP LPAC Physical Dimension**

- 3.9.1. Maximum Allowable Footprint (L x W) Inches: 106 X 66
- 3.9.2. Maximum Allowable Weight (lbs.): N/A.

3.10. **200 HP LPAC Electrical Requirements:**

- 3.10.1. Operating Voltage: 460 VAC 3 Phase 60 Hz
- 3.10.2. Maximum Starting Amperage: 500A
- 3.10.3. Maximum Operating Amperage (FLA): 400A
- 3.10.4. All onboard low voltage controls must have integral transformer to provide necessary voltage stepdown. Compressor must operate from one 460VAC circuit.
- 3.10.5. Equipment shall come with Camlok connections on the unit (J series locking style, i.e.:1016 series)

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3.11. 200 HP LPAC Operating and Safety Controls:

- 3.11.1. Compressor shall cycle using control logic based on maintaining system pressure.
- 3.11.2. Compressor and any pressure vessels within the package must be fitted with appropriate safety relief valve(s).
- 3.11.3. Pressure relief set point shall not exceed the working pressure rating of any component in line with the compressed air system (125 PSI).
- 3.11.4. Alarm(s)/controls to maintain compressor operable and all compressed air quality requirements as defined in this statement of work.
- 3.11.5. The Tooling Air Compressor shall be equipped with Remote Emergency Stop Switch E-stop as defined in paragraph 3.8.11. of this statement of work

3.12. 1000 Gal Compressed Air Receiver Tank Requirements:

- 3.12.1. 1000 Gal, ASME rated Pressure Vessel
- 3.12.2. Minimum working pressure rating: 150 PSIG
- 3.12.3. Pressure Relief valve and tank drain meeting the pressure requirements to protect the supplied air receiver and compressed air system.
- 3.12.4. Tank maintenance & construction shall meet all requirements in accordance with Section 2.1.2.4
- 3.12.5. Vertical mounting Orientation.
- 3.12.6. Tank shall be equipped with lashing/ tie down points and base mount bolt pattern to enable the tank to be properly secured.
- 3.12.7. 3 IN connections: SPUD Fittings, M & F
- 3.12.8. Air Receiving tank shall be 100% free of any oily residue and shall be suitable to support and meet all requirements defined in section 2.1.2.

3.13. Desiccant Dryer Requirements:

- 3.13.1. Nominal Inlet Flow rate: 800 SCFM
- 3.13.2. Compressed air outlet flow Dew Point: -40°F PDP under full load of the supplied tooling air compressor.
- 3.13.3. Ambient Operating Temperature range: -10°F to 110°F
- 3.13.4. Maximum Allowable Footprint (L x W x H) Inches: 105 X 58 x 72
- 3.13.5. Maximum Allowable Weight (lbs.): N/A.
- 3.13.6. Nominal Rated Working Pressure (PSIG): 150
- 3.13.7. Inlet / Outlet Connections: 3IN NPT
 - 3.13.7.1. Note: If the air dryer is manufactured with 2 IN NPT connections, adaptors shall be required to enable 3 IN SPUD connections for hose connections.
 - 3.13.7.2. NPT to 3 IN SPUD adaptors.
- 3.13.8. Dryer shall be equipped with particulate and oil removal pre-filters c/w automatic condensate drains and a one micron dust after filter.
- 3.13.9. The desiccant dryer shall consume no more than 18% of the LPA input volume for purging to a -40°F Dew point..
- 3.13.10. The Desiccant dryer shall operate from one circuit rated at: 120 VAC, single phase, 20 Amp, 60Hz.
- 3.13.11. Dryer shall be UL listed.

3.14. 400 HP Electric LPAC Requirements-[Blast Air System].

- 3.14.1. Nominal Compressed air flow rate: 1550 SCFM
- 3.14.2. Maximum system pressure: 150 PSIG.
- 3.14.3. Regulated delivered Air Pressure Set point: 115 PSIG.
- 3.14.4. Compressor shall be designed to deliver 100% Oil Free air. Oil Flooded compressors are not acceptable.
- 3.14.5. Compressor connection: 3 Inch. Compressor must be supplied with Male/Female SPUD Fitting(s) as required for connection to downstream drying/filtering equipment.

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- 3.14.6. Compressor shall be a fully packaged Air-Cooled unit including intercooler/after cooler to maintain a package maximum discharge compressed air temperature of 100F under all ambient conditions.
- 3.14.7. Compressor shall be enclosed in a weatherproof enclosure suitable for outdoor operation with an ambient operating temperature range of -10°F to 110°F without the need for additional shelter/protection.
- 3.14.8. All opening/connections/flanges are to be cleaned and capped.
- 3.14.9. Compressor shall be UL listed.
- 3.14.10. All vendor supplied equipment shall be capable of operating in below freezing temperatures and include freeze protection measures on components that are prone to freezing (e.g. condensate line, water separators, and drains)
- 3.14.11. Blast Air LPAC must be pre-wired for and compatible with a Remote Emergency Stop (E-stop) Switch. This will enable the LPAC to be shut down in response to an emergency condition where Blast air needs to be secured. The Contractor shall route all E-stops switch & control wiring to a central location designated by the government where the breathing air LPAC equipment can be secured.
 - 3.14.11.1. The E-stop cable distance to be routed is estimated at 500 Ft, however, the Contractor shall be prepared to provide & route up to 1500 FT of wire to accommodate complex routing paths, cable routing around obstacles, service bays, electrical service chases, etc.
 - 3.14.11.2. E-stop switch shall be equipped with protective Plexiglas or acrylic style clear cover to prevent inadvertent activation.
 - 3.14.11.3. E-stop cabling must utilize heavy-duty industrial cable for use in flexible, portable, and extra-hard usage applications (type SO cable or equivalent).
- 3.14.12. All opening/connections/flanges are to be cleaned and capped.

3.15. 400 HP LPAC Physical Dimension

- 3.15.1. Maximum Allowable Footprint (L x W) Inches: 120 X 76
- 3.15.2. Maximum Allowable Weight (lbs.): N/A.

3.16. 400 HP LPAC Electrical Requirements:

- 3.16.1. Operating Voltage: 460 VAC 3 Phase 60 Hz
- 3.16.2. Maximum operating Amperage (FLA): 800A
- 3.16.3. All onboard low voltage controls must have integral transformer to provide necessary voltage stepdown. Compressor must operate from one 460VAC circuit.
- 3.16.4. Equipment shall come with Camlok connections on the unit (J series locking style, i.e.:1016 series)

3.17. 400 HP LPAC Operating and Safety Controls:

- 3.17.1. Compressor shall cycle using control logic based on maintaining system pressure.
- 3.17.2. Compressor and any pressure vessels within the package must be fitted with appropriate safety relief valve(s).
- 3.17.3. Pressure relief set point shall not exceed the working pressure rating of any component in line with the compressed air system (125 PSI).
- 3.17.4. Alarm(s)/controls to maintain compressor operable and all compressed air quality requirements as defined in this statement of work.
- 3.17.5. The Blast Air Compressor shall be equipped with Remote Emergency Stop Switch (E-stop) as defined in paragraph 3.14.11. of this statement of work

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3.18. 1000 Gal Compressed Air Receiver Tank Requirements:

- 3.18.1. 1000Gal, ASME rated Pressure Vessel
- 3.18.2. Pressure relief valve and tank drain meeting the pressure requirements to protect the supplied air receiver and compressed air system.
- 3.18.3. Minimum working pressure rating: 150 PSIG
- 3.18.4. Tank maintenance & construction shall meet all requirements in accordance with Section 2.1.2.4
- 3.18.5. Vertical mounting Orientation.
- 3.18.6. Tank shall be equipped with lashing/ tie down points and base mount bolt pattern to enable the tank to be properly secured.
- 3.18.7. 3 IN connections: SPUD Fittings, M & F

3.19. Desiccant Dryer Requirements:

- 3.19.1. Nominal Inlet Flow rate: 1550 SCFM
- 3.19.2. Nominal Dew Point Operating points: -40°F PDP under full load of the supplied breathing air compressor.
- 3.19.3. Ambient Operating Temperature range: -10°F to 110°F
- 3.19.4. Maximum Allowable Footprint (L x W) Inches: 110 X 80
- 3.19.5. Maximum Allowable Weight (lbs.): N/A
- 3.19.6. Nominal Rated Working Pressure (PSIG): 150
- 3.19.7. Inlet / Outlet Connections: 3IN SPUD
- 3.19.8. Dryer shall be equipped with particulate and oil removal pre-filters c/w automatic condensate drains and a one micron after filter.
 - 3.19.8.1. Post compressor shall be equipped with particulate and coalescing oil removal pre-filters w/ automatic condensate drains and a one micron dust after filter.
- 3.19.9. The desiccant dryer shall consume no more than 18% of the LPA input volume for purging to a -40°F Dew point.
- 3.19.10. The Desiccant dryer shall operate from one circuit rated at: 120 VAC, single phase, 20 Amp, 60Hz.
- 3.19.11. Dryer shall be UL listed.

3.20. Compressed Air Distribution Manifolds:

- 3.20.1. The compressed air system will require a distribution manifold to supply air to various tooling equipment from the receiving air tank.
 - 3.20.1.1. Manifold Input: The distribution manifold shall be supplied from the receiving air tank via a 3 inch inlet hose equipped with SPUD connections.
 - 3.20.1.2. Manifold 3 Inch Flow Through Output: The distribution manifold shall be equipped with one 3 IN outlet connection to enable 3 Inch hoses sections to “flow through” each given manifold, enabling several manifolds to be “daisy chained” in series to a single 3 IN pneumatic distribution line.
 - 3.20.1.3. 3 IN manifold Input and 3 IN manifold output shall be configured as a matched set of MxF SPUD connections.
 - 3.20.1.4. Manifold Branch Outputs: The distribution manifold shall contain 3 IN manifold Input and 3 IN manifold output with six (6) output connections at 1 1/4 IN with pneumatic SPUD connections. Each output shall have an isolation valve so each branch lines may be individually isolated.

3.21. Equipment Air Hose Connection & Distribution Hose Requirements:

- 3.21.1. The contractor shall, on delivery, make all 3 IN hose connections between compressor, dryer, and air receiver, and all hose connections must have safety cables (Safety Restraint Devices) installed to prevent hose whip in case of accidental separation of coupling or clamp device. The Contractor shall provide the necessary length to make all connections.
- 3.21.2. The Contractor shall provide 3 IN, non-collapsible, pneumatic rubber hose- 50 FT. hose length per sections. Connections shall be industry standard MxF SPUD connections.

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- 3.21.3. The Contractor shall provide 2 IN, non-collapsible, pneumatic rubber hose- 50 FT. hose length per sections. Connections shall be industry standard MxF SPUD connections.
- 3.21.4. All compressed air hose and fittings shall have a minimum working pressure of 150 PSI.
- 3.21.5. The Contractor shall provide all necessary interconnecting hose, elbows, & hardware.

3.22. Equipment Start-up/ Commissioning Services

- 3.22.1. Contractor must provide start-up and commissioning services to prepare the equipment to be placed into service. This includes but is not limited to the following:
 - 3.22.1.1. Energizing low pressure compressed air equipment
 - 3.22.1.2. Pressurizing the system
 - 3.22.1.3. Performing leak check of all contractor made connections
 - 3.22.1.4. Adjusting setpoint(s) to meet the requirements of this Statement of Work
 - 3.22.1.5. Verifying satisfactory operation of all drains installed on equipment
 - 3.22.1.6. Troubleshooting/adjudicating any alarm conditions present on start-up (See 6.4)
 - 3.22.1.7. Training Government operators in equipment function, control and alarm(s)/indications.

4. NOTES:

- 4.1. **Construction:** Except as noted for a particular machine type, units shall be a single integrated framed structure manufactured as a complete entity (Fully Packaged) to include all devices, accessories, and fittings necessary to perform the function for which it was designed. Each unit shall be capable of continuous, year-round, outdoor, unmanned operation even at minimal loading while exposed from inclement to extremely harsh weather (e.g. deep snow, strong & gusty winds, and/or heavy rain) and varied ambient conditions (-10 to +110°F & 0 to 100% RH). Machines shall meet State and federal requirements for its type such as the National Electrical Code (NEC) for electrical devices / components and the American Society of Mechanical Engineers (ASME) for pressurized vessels / components. Machines shall be modular in design so as to act as a stand-alone unit; compact so as to fit in the tight confines of a drydock or pier space(s); portable so as to be transported by forklift and then lifted by crane; and ruggedly constructed so as to endure a marine industrial environment. Machines shall be newly manufactured / completely refurbished and factory tested within ten (10) years.
- 4.2. **Electrical:** Electrical components shall conform/meet or exceed a NEMA standard (NEMA 3) that allows for unsheltered outdoor use of the unit. Each machine shall require only one type / source of electrical power (460 VAC +5% / three-phase / 60 hertz or 120 VAC +5% / Single Phase / 60 hertz). The type of electrical power required (voltage & amperage) must be clearly indicated on the outside of the equipment, readily detected, and collocated with all other electrical information. The Contractor shall provide the Contracting Officer's Representatives (COR) and Technical Points of Contact (TPOC) written notification of the following details: (i) electrical power type (amperage & voltage); (ii) type of electrical connectors shall be delivered with fabricated, watertight, Camlok connections (iii) breaker type/size or recommended capacity for the electrical source; and (iv) whether the machine is furnished with over-current protection at time of delivery to allow for safe connection to Government provided power source. This information must be included in technical submittals for review by engineering to allow for planning efforts by the project.
- 4.3. **Contractor Identification Of Itemized And Ancillary Equipment:** The Contractor shall uniquely tag and permanently mark all of its ancillary and itemized equipment / hardware (e.g. manifolds, hose assemblies, pipe fittings, duct adapters, ducting, etc.) so as to differentiate from that owned by either the Shipyard or other Contractors.
- 4.4. **Dead Front Control Panel or External Controller:** Access to basic machinery controls, such as a Dead Front Control Panel or External Controller, shall be such that there is non-exposure to energized live electrical circuits or other components regardless of voltage, to include control circuitry, by a person on the operating side of the equipment.

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5. **GOVERNMENT FURNISHED MATERIAL (GFM):** The Government will provide the equipment listed below, while operating the equipment and will make fork lifts and operators available to load and unload the equipment from the Contractor's vehicles.

- 5.1. Breathing Air Purifier.
- 5.2. CO alarm & monitoring equipment

6. **RESPONSIBILITIES.**

6.1. **Government**

- 6.1.1. Equipment Technical Evaluation: C230 Engineering
- 6.1.2. Receipt Inspection: COR, TPOC & Zone Manager (ZM)
- 6.1.3. Schedule equipment delivery & returns: COR, TPOC & ZM
- 6.1.4. Schedule equipment service, repair, and training: COR, TPOC & Project Zone Manager.

6.2. **Contractor**

- 6.2.1. Any and all required State approved safety features (i.e. safety pressure relief valves, low water cutout, over pressure protection) installed with applicable State certification.
- 6.2.2. The contractor shall maintain a stringent maintenance program and be responsible for all preventative maintenance, except for daily check of liquid levels. The contractor shall provide oil, filters, coolant, and lubricant for engine and compressor operation during the rental period.
- 6.2.3. The contractor shall provide repair service within twelve (12) hours after notification by the Contracting Officer or his/her representative. In the event the machine cannot be returned to service within twenty-four (24) hours of notification, the Contractor shall deliver a replacement at no extra charge to the Government.
- 6.2.4. The contractor shall provide equipment start-up, and on site commissioning per SOW 3.22.
- 6.2.5. The contractor shall be responsible for equipment failure not arising out of abuse or negligence by the government.
- 6.2.6. There shall be no changes unless authorized by the Contracting Officer.
- 6.2.7. The Contractor shall provide technical support to PNSY project personnel as required.
- 6.2.8. The Contractor shall provide all deliverables specified in section 6.5.

6.3. **Maintenance**

- 6.3.1. **Surveillance Checks.** The Government will conduct daily surveillance checks of the equipment, which is limited to a visual inspection verifying no alarm conditions or obvious deficiencies are present (Ex. Abnormal Noises, Vibrations, leaks). Except for daily surveillance checks, the Contractor is responsible for all equipment preventative maintenance, weekly inspections and repairs for the duration of the lease. The Contractor shall coordinate with the COR and TPOC, in writing, to schedule preventative maintenance and weekly inspections.

- 6.3.2. **Preventative Maintenance & Weekly Inspections.** The Contractor shall perform Preventative Maintenance and Weekly Inspections of the equipment during the contract. The Contractor shall provide a Preventative Maintenance (PM) plan with schedules which are equipment specific and include a list of actions for both weekly inspections and PM. At a minimum, the list of actions shall meet all requirements recommended by the Original Equipment Manufacturer (OEM). If a list of actions is not available from or recommended by the OEM, the Contractor shall develop a list of actions and perform these services in accordance with the SOW. Weekly inspections shall be performed by a qualified service technician provided by the Contractor and shall consist of checking all functional operating parameters of each individual piece of equipment including, but is not limited to: verifying correct equipment set-up, performance and operation, checking for diagnostic or trouble codes, and verifying amperage of all electric heating coils. The PM plan actions list for each piece of equipment shall include a comprehensive list of the components to be serviced as well as identifying

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the service frequency (e.g., all consumables to be replaced, bearing grease points, belt replacement / tensioning, filter changes, refrigerant charge, oil changes, other lubrication, coolant refills, liquid top-offs, etc.) The Contractor shall prepare and submit a technical field report that identifies what was included in the weekly inspection, the results of the inspection, and which PM plan actions were performed, what system, equipment or component was serviced, when the PM was performed and next scheduled PM date. Copies of all reports shall be delivered to the COR and TPOC upon completion of the service.

6.3.3. **Repairs.** The Contractor shall provide a qualified service technician to repair all equipment deficiencies and malfunctions and return the system to an operable condition. The response time for the repair or replacement of deficient equipment shall meet the requirements defined in section 6.2.

6.4. **Training.** The Contractor is responsible for and shall provide training sessions to Shipyard Personnel on the operation and setup of all equipment. Training shall be available throughout the duration of the lease to accommodate equipment swap out and seasonal equipment operation.

6.5. **Deliverables.** The Contractor shall deliver the following information, electronically to the COR and TPOC, and in hard copy when the equipment is delivered to the Government:

6.5.1. Technical Instructions (TIs) such as Operations & Maintenance Manuals, Operating Instructions, Controls Manual, and/or Mechanical/Electrical Manual with Schematics; Note, all Technical Instructions shall be located in a water proof case affixed to each machine.

6.5.2. Points of contact with telephone numbers for requesting emergency repairs 24 hours 7 days a week

6.5.3. Preventative Maintenance Schedules. (See section 6.3)

6.5.4. Equipment Data Sheet (EDS)

7. DELIVERY / ACCEPTANCE / RETURN.

7.1. **Delivery.** Except as noted below in paragraph 7.3, the Contractor shall deliver machinery that is fully-charged / fueled, fully-assembled, ready-to-use, in a first-class-operating condition and functions as intended. The machinery with its support equipment shall be delivered to the site on or prior to the request date. The Contractor shall confirm the date and time of delivery prior to each delivery with the COR and TPOC, if appointed. The Government can provide forklift services for removing machinery from Contractor's transport during work hours. The Contractor shall provide the COR and TPOC with an itemized list of all equipment delivered at the time of delivery.

7.2. **Acceptance.** Upon satisfactory completion of inspection and testing of the equipment, the authorized Government Representative (Portsmouth Naval Shipyard, Code 420 COR) will sign and forward the acceptance document(s) to Code 220 Production Control for completion. Unloading of equipment from Contractor's transport does not signify acceptance.

7.3. **Start Up-** Upon delivery, the Contractor shall: **(i)** provide start up and commissioning of the equipment to verify its operability per SOW section 3.22; **(ii)** provide all deliverables as required per SOW section 6.5 ; and **(iii)** train designated Shipyard personnel on how to properly run the equipment to include its operation, control, daily checks, & service contacts per SOW section 6.4.

7.4. **Security-** To be allowed onto a government Shipyard, personnel must be US citizens and in possession of appropriate personal identification. Proof of US citizenship may be necessary and can be required for access to the equipment installation location. Allowable documentation of US citizenship is an official government issued Certificate of Birth or Naturalization (w/ raised seal & not a photocopy) or a US passport (can be expired). Vehicular access requires a non-expired personal Driver's License and appropriate vehicle

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identification or registration. The transport vehicle must have the company's name / logo prominently displayed. To gain access to the drydock environs, personnel must be escorted and possess / wear their own personal safety gear.

7.5. **Receipt Inspection Criteria-** Equipment must meet all criteria defined in this specification.

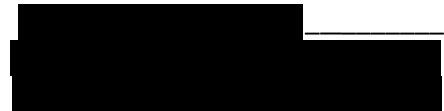
7.6. **Final Acceptance.** Upon satisfactory completion of inspection and test of the equipment, the authorized Government Representative (Portsmouth Naval Shipyard Code 990 Equipment Manager or Project Temporary Service Zone Manager) will sign and forward the acceptance document(s) to the POC as designated on the Contract and/or the COR for completion. Unloading of equipment from Contractor's transport does not signify acceptance.

7.7. **Return.** The Government will provide a temporary storage location convenient for Contractor pick-up. The Government will notify the Contractor as well as the Contracting Office as to when the rental Item has been released and is available for pick-up. The last day of the rental period shall be the next normal workday after the Contractor has been so notified. The Contractor shall be responsible for shipping. The Government can provide forklift services during work hours for placing the equipment onto the Contractor's transport.

8. ATTACHMENTS

8.1. N/A

ENGINEERING APPROVAL

A large black rectangular redaction box covering the signature area of the Engineering Approval section.